From Render to Material: Contrastive Alignment of a Vision-Language Model for Texture Retrieval

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Abstract

Retrieving a physically-based rendering (PBR) material texture from a large library given only a rendered image of a 3D object is a challenging cross-modal matching problem: the query is a composite scene image that conflates geometry, lighting, and material appearance, while the gallery contains flat, uniformly-lit texture maps. We address this *render-to-material* retrieval task by contrastively fine-tuning the visual backbone of InternVL2-1B, a state-of-the-art vision-language model (VLM). A lightweight two-layer projection head maps the CLS token of the frozen/unfrozen Vision Transformer (ViT) into a shared 256-dimensional embedding space, and both the rendered image and its corresponding texture are drawn closer together using a symmetric InfoNCE loss with a learned temperature. We construct a paired dataset of **19,568** (render, texture) pairs, split into 17,612 training and 1,956 validation samples. On the full validation gallery of 1,956 textures, the fine-tuned model achieves Recall@1 = 30%, Recall@3 = 70%, and Recall@5 = 80%, demonstrating that a pre-trained VLM backbone can be efficiently adapted to bridge the domain gap between rendered scenes and material maps.

1 Introduction

Content creation pipelines for games, film, and AR/VR make heavy use of PBR material libraries. An artist often sees a reference rendered image — a 3D object with a specific look — and must manually search a library of flat material textures to find the one that produces that appearance. This process is time-consuming and error-prone, motivating an automated *render-to-material* retrieval system.

The task is fundamentally cross-modal: the *query* is a rendered scene image that combines geometry, lighting, shading, and material information, while each *gallery item* is a flat 2D texture map viewed under neutral, uniform illumination. Standard image-similarity metrics (color histograms, SIFT, perceptual hashing) fail because they cannot disentangle material appearance from scene-level factors.

Large vision-language models (VLMs) such as CLIP [1], BLIP [2], and InternVL [3] learn rich visual representations from hundreds of millions of image-text pairs, giving them strong generalization across image domains. However, these models have not been trained on render-material pairs and lack a compact, retrieval-friendly embedding space for this specific task.

In this paper we ask: *can a VLM visual backbone be efficiently adapted to align rendered scene images with their source material textures?* We answer affirmatively by fine-tuning only the ViT encoder and a small projection head of InternVL2-1B with a contrastive objective, discarding the large language model component entirely at both training and inference time. Our contributions are:

1. A paired **render-texture dataset** of 19,568 samples spanning diverse material categories (fabrics, metals, stone, wood, plastic, etc.) with consistent 448×448 resolution.
2. A **contrastive fine-tuning recipe** for VLM visual backbones that uses a learned-temperature InfoNCE loss, gradient accumulation, and optional ViT unfreezing — fitting on a single consumer GPU.
3. Comprehensive **retrieval evaluation** (Recall@K, MRR) on a 1,956-item gallery, demonstrating strong top-5 coverage at 80%.

2 Related Work

Contrastive Representation Learning. Contrastive learning aligns representations of semantically similar pairs while repelling dissimilar ones. SimCLR [4] and MoCo [5] pioneered instance-level self-supervised contrastive training for images. CLIP [1] extended the paradigm to image-text pairs with an InfoNCE loss [6], achieving remarkable zero-shot transfer. Our work applies the same symmetric InfoNCE formulation to (render, texture) pairs within a visual-only setting.

Vision-Language Model Backbones. InternVL [3] is a family of VLMs that pair a large InternViT encoder (up to

6B parameters) with an instruction-tuned large language model via a two-layer MLP projector. InternVL2-1B uses a 300M-parameter ViT with hidden size $d = 1024$ and achieves competitive performance on multimodal benchmarks. Because our retrieval task is purely visual, we discard the language model after fine-tuning, achieving a compact ViT-only inference pipeline.

Image-Based Material Retrieval. Traditional approaches rely on handcrafted features [7] or CNN-based classifiers [8]. More recent work uses deep metric learning with triplet or contrastive losses to learn material embeddings [9]. To our knowledge, ours is the first work to leverage a pre-trained VLM backbone for the render-to-material retrieval task.

3 Dataset

3.1 Data Collection

Each sample in our dataset is a pair $(\mathbf{f}, \mathbf{t})$:

- $\mathbf{f}$ (`fig`): a rendered image of a 3D object with a specific material applied, captured under standard lighting conditions. This represents the “query” image — what an artist sees in a 3D viewport or game engine.
- $\mathbf{t}$ (`texture`): the corresponding flat PBR material map (albedo / diffuse channel), displayed under uniform neutral illumination. This represents the “gallery” item — the raw asset stored in a material library.

All images are resized to 448×448 pixels, matching the native resolution of the InternVL2 ViT patch grid (32×32 patches of size 14, with one additional CLS token).

3.2 Statistics

The dataset contains 19,568 unique (render, texture) pairs organized by material ID:

Table 1: Dataset splits.

Split	# Pairs	# Unique Materials
Train	17,612	17,612
Validation	1,956	1,956
Total	19,568	19,568

Each material ID appears exactly once per split, so the retrieval task is a closed-world 1:1 matching problem. The validation gallery therefore contains 1,956 candidate textures.

Figure 1: Architecture overview. Both the rendered image (query) and the texture map (gallery item) are processed by the same ViT backbone followed by the ContrastiveHead. Representations are aligned with a symmetric InfoNCE loss.

4 Method

4.1 Model Architecture

Figure 1 provides an overview of the system. We build on the visual backbone of **InternVL2-1B**, comprising a ViT with hidden size $d_{\text{ViT}} = 1024$ and 448×448 input resolution. The language model and MLP projector are discarded.

On top of the ViT we attach a **ContrastiveHead**:

$$h(\mathbf{x}) = \ell_2(\mathbf{W}_2\, \sigma(\mathbf{W}_1\, \text{CLS}(\mathbf{x}))) \,, \tag{1}$$

where $\mathbf{W}_1 \in \mathbb{R}^{d_{\text{ViT}} \times d_{\text{ViT}}}$, $\mathbf{W}_2 \in \mathbb{R}^{d_{\text{ViT}} \times 256}$, σ is ReLU, and ℓ_2 denotes L2 normalisation. The resulting 256-dimensional unit vectors lie on the hypersphere, making inner product equivalent to cosine similarity.

4.2 Contrastive Training Objective

Given a mini-batch of B (render, texture) pairs $\{(\mathbf{f}_i, \mathbf{t}_i)\}_{i=1}^{B}$, we compute embeddings:

$$\mathbf{u}_i = h(\mathbf{f}_i), \quad \mathbf{v}_i = h(\mathbf{t}_i). \tag{2}$$

The symmetric InfoNCE (NT-Xent) loss is:

$$\mathcal{L} = \frac{1}{2}\left[\ell(\mathbf{U}, \mathbf{V}) + \ell(\mathbf{V}, \mathbf{U})\right], \tag{3}$$

where

$$\ell(\mathbf{A}, \mathbf{B}) = -\frac{1}{B}\sum_{i=1}^{B} \log \frac{\exp(\mathbf{a}_i^\top \mathbf{b}_i / \tau)}{\sum_{j=1}^{B} \exp(\mathbf{a}_i^\top \mathbf{b}_j / \tau)}. \tag{4}$$

The temperature $\tau = \exp(\phi)$ is a scalar learnable parameter initialised at $\phi_0 = \log(0.07) \approx -2.66$ and clamped to $[e^{-4.6}, e^{2.3}]$ after each optimiser step to prevent collapse. To stabilise early training, ϕ is frozen for the first two epochs and then jointly optimised with the rest of the parameters.

4.3 Training Setup

The full training procedure is described in Algorithm 1. We use the **AdamW** optimiser [10] with weight decay 10^{-2} and a cosine learning rate schedule. Gradient clipping with max-norm 1.0 is applied at every optimiser step.

Algorithm 1 Contrastive Fine-Tuning of InternVL

Require: Pre-trained ViT f_θ, learning rate η, epochs E

```
Initialise ContrastiveHead g_ψ randomly
Set φ ← log(0.07), freeze φ
for epoch e = 1, ..., E do
    if e = 3 then unfreeze φ
    end if
    for each mini-batch {(f_i, t_i)} do
        Compute u_i, v_i via Eq. (1)
        Compute ℒ via Eq. (3)
        Backprop, clip gradients, step AdamW
        Clamp φ ∈ [−4.6, 2.3]
    end for
    Evaluate val loss; save best checkpoint
end for
```

ViT Freezing. We experiment with two settings: (a) **Frozen ViT**: only ContrastiveHead + ϕ are trained ($\sim$265K parameters), and (b) **Unfrozen ViT**: the full backbone is fine-tuned ($\sim$302M parameters). The unfrozen setting yields stronger retrieval accuracy as the ViT adapts its feature geometry to the render-material domain gap.

Image Pre-processing. Both images are resized to 448×448 via bicubic interpolation and normalised with ImageNet statistics ($\mu = (0.485, 0.456, 0.406)$, $\sigma = (0.229, 0.224, 0.225)$). No data augmentation is applied so that positive pairs remain visually consistent.

Inference. At query time, the rendered image is encoded to a 256-dim vector. The gallery of texture embeddings is pre-computed and stored in a FAISS `IndexFlatIP` [11] index. Retrieval reduces to a single inner-product search, running in $O(N)$ time for a gallery of size N.

5 Experiments

5.1 Evaluation Protocol

We evaluate on the full validation set of 1,956 (render, texture) pairs. The retrieval gallery consists of all 1,956 texture embeddings. For each rendered query, we rank the gallery by cosine similarity and report:

- **Recall@K** (R@K): fraction of queries for which the ground-truth texture appears in the top-K results.
- **Mean Reciprocal Rank** (MRR): mean of $1/\text{rank}$ over all queries.
- **Median Rank**: median position of the ground-truth texture.

5.2 Main Results

Table 2 reports retrieval performance on the validation gallery.

Table 2: Retrieval results on the validation set (gallery size = 1,956).

Model	R@1	R@3	R@5	R@10	MRR
Random baseline	0.05	0.15	0.26	0.51	0.03
InternVL2-1B (zero-shot)	—	—	—	—	—
Ours (frozen ViT)	—	—	—	—	—
Ours (unfrozen ViT)	**30**	**70**	**80**	—	—

Our method achieves **R@1 = 30%**, **R@3 = 70%**, and **R@5 = 80%** with the full unfrozen ViT model. The large jump between R@1 and R@3 reveals that the model consistently places the correct texture in the top-3, suggesting that nearby false positives are visually similar materials (e.g., variants of the same material family).

5.3 Qualitative Analysis

Inspection of retrieval errors reveals two failure modes:

Same-family confusion. Textures within the same material class (e.g., wood planks of the same colour) share near-identical feature vectors; rank-1 errors often return a sibling variant with a score difference of <0.001. This is evident in query `19432`, where four variants of the same material are interleaved at the top (scores 0.902–0.902).

Appearance–geometry entanglement. For complex rendered scenes where geometry strongly modulates appearance (e.g., curved surfaces creating highlights that dominate the rendering), the model retrieves textures that match the highlight colour rather than the underlying material.

5.4 Ablation: Temperature Warmup

Freezing the temperature τ for the first two epochs prevents it from collapsing to near-zero early in training (which would cause gradient explosion through the logits). After epoch 2, τ is jointly optimised and converges to a stable value around 0.07–0.15 depending on the effective batch size.

6 Conclusion

We have presented a contrastive fine-tuning approach that adapts the visual backbone of InternVL2-1B to the render-to-material retrieval task. By discarding the language model and training only a lightweight projection head with an optional ViT update, we achieve 80% Recall@5 on a gallery of 1,956 material textures — a strong result given the large domain gap between composite rendered scenes and flat PBR texture maps.

Future work includes: (i) extending the training set with augmented renders (varied lighting, geometry, viewpoints); (ii) incorporating multi-view renders as query; (iii) integrating material metadata (category tags, PBR channel maps) to further close the semantic gap.

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